6-1 Analog and Digital

Point!

Analog and Digital

Analog



Digital



(1) ([1]**Analog**) : Refers to quantities that can be measured continuously and with fine precision, such as mass, time, or temperature. Data represented in analog form is referred to as ([2]**analog data**).

(2) ([3]**Digital**) : Refers to quantities that continuously change and are represented numerically by segmenting at regular intervals. Data represented in digital form is referred to as ([4]**digital data**).

(3) ([5]**A/D conversion (digitization)**) : The process of converting analog data into digital data.

(4) ([6]**D/A conversion**) : Converting digital data into analog data.

(5) Advantages of Digital Data

[1] Data can be ([7]**accumulated**) or ([8]**duplicated**) without degradation.

[2] It is easy to ([9]**modify**) and ([10]**edit**) the data.

[3] It is possible to efficiently ([11]**communicate**) the information.

[4] It is possible to ([12]**combine**) various types of media.

(6) ([13]**Binary**) : A method of representing numbers using two types of digits, 0 and 1.

- In computers, all types of information, such as numbers, characters, audio, and images, are represented in ([14]**binary**).

Warm Up

For each of the following statements **A** to **D**, answer whether the statement describes analog or digital.

A A thermometer that represents a continuous quantity using the length of a mercury column.

B Accurate and highly reproducible, without any degradation due to duplication or transmission.

C A quantity that changes continuously can be represented using the numbers 0 and 1.

D All information (numbers, text, audio, images, videos, etc.) can be represented in binary.

Explanation

A: Analog

B: Digital

C: Digital

D: Digital

Try

Answer the following questions.

(1) Choose the terms that best fit into the blanks [1] to [5] from the options **A** to **F** below.

> A quantity that is commonly found in our daily lives and changes continuously, such as time or temperature, is called ([1]). In contrast, a quantity that represents a continuously changing value by dividing it into regular intervals and expressing it numerically is called ([2]). Converting analog data to digital data is called ([3]) conversion, while the reverse is called ([4]) conversion.
>
> Furthermore, when handling characters, images, audio, video, and other data on a computer, the data is represented using two numbers, 0 and 1, which is referred to as ([5]).

A Binary **B** Decimal **C** Digital **D** Analog
E A/D **F** D/A

(2) For each of the following statements **A** to **E**, answer whether the statement describes analog or digital.

A Accumulation and duplication of information can be easily achieved.
B Temperature is measured by assessing the height of the mercury column.
C Even if some distortion occurs in the electrical signal due to noise during transmission, it can be easily corrected.
D Quantities that change continuously, such as time or temperature.
E It is possible to comprehensively handle various forms of media.

Exercise

1 Answer the following questions.

(1) What is the term for quantities that can be measured continuously and with fine precision, such as mass or time?
(2) What is the term for the process of converting digital data into analog data?
(3) What is A/D conversion? Explain briefly using the terms “digital” and “analog.”
(4) What is the term for the method of representing numbers using two types of digits (0 and 1)?

2 Choose one statement that is incorrect in regard to digital data from the options **A** to **D** below, and answer using the letter.

A Accumulation and duplication of information can be easily achieved.
B It is possible to efficiently communicate the information.
C It can express quantities that change continuously, such as time and temperature.
D It is possible to combine various types of media.